

SHUVAM PAL

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Education

Dayananda Sagar College of Engineering

Bengaluru, India

B.E. Artificial Intelligence and Machine Learning

Experience

Bentham AI 

Dec 2025 – Feb 2026

Product Engineering Intern: Node.js, JavaScript, Go, Docker, GCP, GitHub Actions

Remote

- Built production **backend automation** with Node.js services, public-site API lookups, browser execution, **retries**, **session recovery**, and **state validation**.
- Kept long-running workflows **recoverable and inspectable**, reducing manual filing time by **85%** across high-friction operational steps.
- Shipped **Dockerized services** through **Google Cloud Run** and **GitHub Actions**, improving deployment velocity by **70%**.

Google Summer of Code 

May 2025 – Sept 2025

Mentee at Invesalio: Python, VTK, wxPython, NumPy, PyTorch, ONNX

Remote

- Built **Python tooling** inside a mature open-source application, working across large data paths, generated artifacts, model outputs, and user-facing inspection workflows.
- Reduced runtime by **87%** with **async/parallel execution** and debugged correctness issues across NumPy, PyTorch/ONNX-style outputs, label mapping, and UI integration.
- Adapted changes to existing architecture, maintainer feedback, and reliability constraints rather than treating the codebase as a greenfield project.

Open Source

[Contributions](#) 

- Contributed to **Kuadrant MCP Gateway** backend-outage grace handling, **Kyverno** policy/infrastructure paths, and IOOS/Tiled data or visualization tooling under maintainer review.

Projects

Hyoka | *Python, FastAPI, PostgreSQL, SQLite, SQLAlchemy, OpenTelemetry, Docker*

- Built a **Python/FastAPI backend platform** for **trace ingestion**, **eval/replay workers**, validation runs, **release gates**, promotion state, and **audit logs**.
- Implemented **SQLAlchemy/Alembic** models, **PostgreSQL/SQLite** storage paths, project API keys, **worker leases**, signed manifests, and reproducible runs around immutable inputs.

Seaweed: Contest Platform | *Go, TypeScript, React, PostgreSQL, Redis, Judge0, Docker, Kubernetes*

[GitHub](#) 

- Built product and platform paths for registration, submissions, sandboxed code execution, admin review, live rankings, and durable result records across contest traffic.
- Designed async verdict polling, PostgreSQL result schemas, Redis-backed runtime state, Docker/Kubernetes deployment, and **sub-5s P95 verdict latency**.

Omni-SSM Voice Agent | *Python, PyTorch, ONNX Runtime, Sherpa-ONNX, Moonshine, Piper*

- Built a supporting **speech/AI engineering project** with STT/TTS experiments, Moonshine Hindi ASR tuning, ONNX Runtime, Sherpa-ONNX, Piper TTS, and provider-profile checks.
- Tracked WER, first-audio latency, memory usage, manifest-driven datasets, and partial fine-tuning runs so model behavior stayed measurable and debuggable.

Technical Skills

Languages: Python, SQL, TypeScript, JavaScript, Go

Frameworks: FastAPI, SQLAlchemy, React, Next.js, Node.js, PyTorch, ONNX Runtime

Tools: PostgreSQL/pgvector, SQLite, Redis, Docker, Kubernetes, GCP, GitHub Actions, OpenTelemetry, Langfuse, Judge0